Establishment of a Real-World Intelligent 5G Network Slicing and Mobility Research Testbed

Project Proposal

June 25, 2026

Contents

1 Executive Summary

This document presents the preliminary requirements for establishing a real-world 5G Network Slicing and Mobility Research Testbed intended to support the development and evaluation of dynamic DNN-based slice switching, mobility management, and intelligent handover mechanisms in a private 5G environment.

The proposed work extends beyond software-based simulation and requires a physical testbed consisting of a 5G Standalone (SA) Core Network, SDR-based Radio Access Network (RAN), commercial 5G User Equipments (UEs), edge computing infrastructure, monitoring platforms, and supporting networking equipment. The objective is to enable experimental validation of slice-aware mobility scenarios under realistic operating conditions and to provide a foundation for future research in AI-assisted networking, Multi-access Edge Computing (MEC), autonomous network management, and emerging 6G technologies.

This document serves as an initial technical assessment of the infrastructure, software, manpower, and research requirements necessary for the successful execution of the project. It identifies the major hardware and software components required for deployment, outlines potential procurement sources, presents indicative budget estimates, and defines the technical competencies expected from project personnel.

A key outcome of this assessment is the identification of the skills and responsibilities required for the appointment of a Junior Research Fellow (JRF) who will be responsible for supporting the deployment, integration, operation, and experimental evaluation of the proposed testbed. The JRF requirements outlined in this document are derived directly from the technologies, tools, and research activities associated with the project.

The information presented herein is intended to support planning, resource allocation, procurement decisions, manpower recruitment, and subsequent preparation of detailed project execution and deployment plans.

2 Problem Statement

Network slicing is a key capability of 5G networks, enabling multiple logical networks with distinct Quality of Service (QoS) requirements to operate over a shared physical infrastructure. While current mobility management mechanisms effectively maintain radio connectivity, they are not designed to optimize slice continuity, application-level performance, edge service availability, or dynamic service requirements during mobility events. As users move across coverage regions, this can result in increased latency, packet loss, service interruption, and inefficient resource utilization.

The objective of this project is to establish a real-world 5G Standalone testbed capable of evaluating dynamic DNN-based slice switching, slice-aware mobility management, and intelligent handover mechanisms under realistic operating conditions. The testbed will provide the infrastructure necessary to study mobility-driven slice selection, QoS preservation, edge computing integration, and future AI-assisted network optimization, while also serving as a foundation for subsequent research activities, manpower development, and advanced networking experimentation.

3 State of the Art

Open5GS, srsRAN, and OpenAirInterface have emerged as the most widely adopted open-source platforms for private 5G research and prototyping. Open5GS provides a standards-compliant 5G Core implementation, while srsRAN and OpenAirInterface enable SDR-based deployment of 5G Radio Access Networks. These platforms are extensively used by universities, research laboratories, and industry teams for evaluating mobility management, network slicing, and edge computing architectures.

The proposed testbed will leverage these technologies to build a real-world private 5G environment capable of supporting dynamic slice switching and mobility experimentation. Their open architecture, active development communities, and compatibility with commercial SDR hardware make them suitable for long-term research and future expansion toward AI-driven networking and 6G studies.

4 Proposed Testbed Overview

The proposed testbed will be established as a multi-layer private 5G research infrastructure consisting of Core Network, Radio Access Network (RAN), Edge Computing, Artificial Intelligence, Monitoring, and User Equipment subsystems. Each subsystem will serve a specific role in enabling the evaluation of dynamic network slice switching, mobility management, and intelligent handover mechanisms under realistic operating conditions.

The Core Network layer will be responsible for subscriber management, session establishment, policy enforcement, and network slicing functions. This layer is expected to be deployed on enterprise-grade rack servers running a 5G Core platform such as Open5GS or a commercial private 5G core solution. The Radio Access Network layer will comprise SDR-based gNodeBs implemented using platforms such as USRP B210/X310 with srsRAN or OpenAirInterface, enabling full control over radio and mobility procedures. Commercial 5G smartphones supporting Standalone (SA) operation will be used as User Equipments (UEs) for real-world mobility and handover experiments.

To support low-latency services and future research activities, a dedicated Multi-access Edge Computing (MEC) infrastructure will be deployed using high-performance compute servers capable of hosting containerized applications and edge services. An AI and Analytics layer consisting of GPU-enabled workstations will be utilized for mobility prediction, traffic analysis, QoS forecasting, anomaly detection, and autonomous network optimization. A centralized monitoring platform based on Grafana, Prometheus, and network telemetry tools will provide visibility into network performance, slice utilization, handover events, and system health. Together, these components will form a scalable research platform capable of supporting current project objectives as well as future investigations into AI-native networking, autonomous network management, and emerging 6G technologies.

5 Testbed Components and Infrastructure Requirements

Table 1: Proposed Testbed Components and Infrastructure Requirements

Subsystem	**Purpose**	**Hardware**	**Software**	**Cost (INR)**
				5G Core Network
Subscriber and session management	Dell PowerEdge / HPE ProLiant Server	Open5GS	3,00,000 – 4,00,000 Radio Access Network	5G gNodeB deployment
2 × USRP B210/X310 SDRs	srsRAN / OAI	5,00,000 – 6,50,000 User Equipment Layer	Mobility and handover testing	4–6 Commercial 5G Smartphones
Android 5G SA	1,00,000 – 2,00,000 MEC Infrastructure	Edge application hosting	Dedicated Edge Server	Docker, Kubernetes
2,00,000 – 3,00,000 AI and Analytics Layer	Mobility prediction and optimization	GPU Workstation	PyTorch, TensorFlow	2,50,000 – 4,00,000 Monitoring Infrastructure
QoS and telemetry monitoring	Monitoring Server	Grafana, Prometheus	50,000 – 1,00,000 Storage Infrastructure	Dataset and log storage
NAS Storage	TrueNAS	75,000 – 1,50,000 Networking Infrastructure	Interconnection of components	Managed Switches and Router
Enterprise Networking Stack 1,00,000 – 1,50,000	50,000 – 1,00,000 Power Infrastructure	Power backup	5 KVA Online UPS	N/A

6 Detailed Architecture

Proposed Real-World 5G Network Slicing and Mobility Research Testbed Architecture

The proposed architecture is organized into six logical layers: User Equipment Layer, Radio Access Network Layer, Core Network Layer, Edge Computing Layer, Intelligence

Layer, and Analytics Layer. Commercial 5G User Equipments communicate with SDR-based gNodeBs deployed using platforms such as USRP B210/X310. The Radio Access Network interfaces with the 5G Core Network, comprising AMF, SMF, UPF, and subscriber management services. A dedicated MEC infrastructure hosts latency-sensitive applications, while an AI-driven controller performs mobility prediction, slice selection, and handover optimization. A centralized monitoring platform continuously collects network telemetry, QoS metrics, and operational statistics to support experimentation and performance evaluation.

6.1 Physical Deployment Architecture

6.2 Physical Deployment Architecture

Proposed Physical Deployment Architecture of the 5G Research Testbed

The proposed deployment consists of a centralized 5G Core Server connected to a managed enterprise switch that interconnects the Radio Access Network, MEC infrastructure, AI platform, monitoring systems, and storage resources. Two SDR-based gNodeBs provide radio coverage for multiple commercial 5G User Equipments, enabling mobility and handover experiments. Dedicated MEC, AI, and monitoring servers support edge computing, intelligent mobility management, analytics, and long-term dataset collection, while the NAS infrastructure provides centralized storage for experimental data and research artifacts.

7 Detailed Architecture

8 SDR and Radio Infrastructure

The Radio Access Network (RAN) shall be implemented using Software Defined Radio (SDR) platforms. SDRs provide flexibility for implementing and modifying physical layer and MAC layer functionality while remaining significantly more economical than commercial telecom-grade base stations.

The proposed deployment will initially utilize two SDR-based gNodeBs supporting 5G Standalone operation.

Table 2: SDR Platform Comparison

Platform	Frequency Range	Cost (INR)	Recommendation
USRP B210	70 MHz – 6 GHz	2.2L-2.8L	Recommended
USRP X310	DC – 6 GHz	10L-15L	Advanced Research
LimeSDR	100 kHz – 3.8 GHz	0.6L-0.8L	Entry Level
BladeRF 2.0	47 MHz – 6 GHz	1.2L-1.8L	Alternative

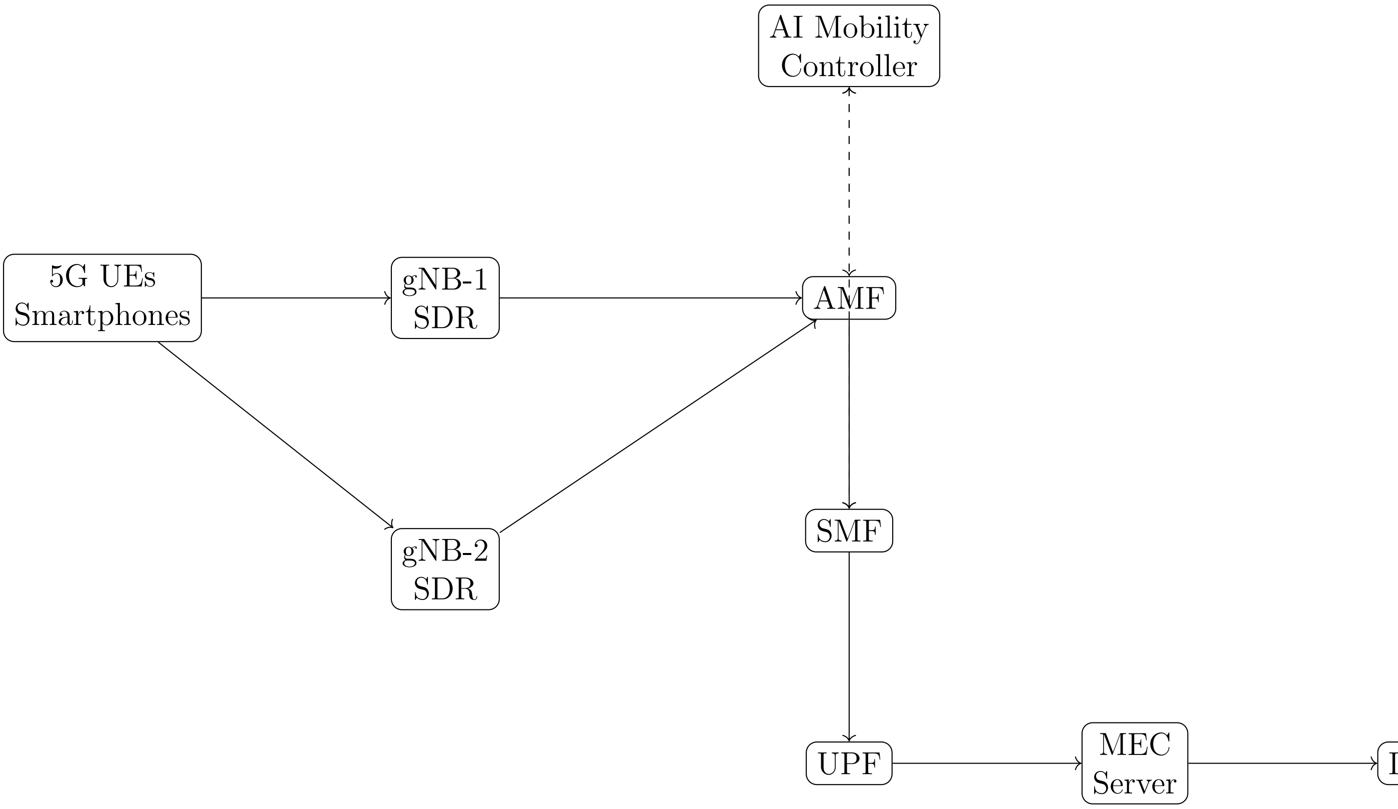

Figure 1: Proposed Real-World 5G Testbed Architecture

9 Multi-access Edge Computing (MEC) Architecture

MEC infrastructure enables application services to be deployed at the network edge, reducing end-to-end latency and improving service responsiveness.

The MEC platform shall host:

- Video Analytics Applications
- AI Inference Engines
- Autonomous Mobility Services
- Smart Campus Applications
- Real-Time Monitoring Services

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UE
|
gNB
|
UPF
|
MEC Cluster
|
Cloud Services
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10 Open5GS and srsRAN Integration

The proposed implementation utilizes Open5GS as the 5G Core Network and srsRAN as the Radio Access Network stack.

Open5GS Components:

- AMF
- SMF
- UPF
- NRF
- UDM
- AUSF
- PCF

srsRAN Components:

- gNB
- Radio Scheduler
- PHY Layer
- MAC Layer
- RLC Layer

11 RF Laboratory Design

Real-world radio experimentation requires a controlled RF environment to prevent interference and ensure repeatable measurements.

Table 3: RF Laboratory Components

Component	Purpose
SDR Rack	Radio Equipment
RF Attenuators	Signal Control
Shielding Cabinet	Isolation
Spectrum Analyzer	RF Measurements
UPS	Power Backup
Monitoring Station	Analytics

12 Spectrum and Regulatory Considerations

The proposed testbed will operate in compliance with applicable regulations issued by the Department of Telecommunications (DoT) and the Wireless Planning and Coordination (WPC) Wing of India.

The testbed will initially operate within a shielded laboratory environment using SDR-based transmission systems and controlled RF attenuation.

No commercial public service shall be provided using the infrastructure.

Potential deployment approaches include:

1. Fully Shielded Laboratory Operation
2. Conducted RF Testing
3. Experimental Licensing Through WPC
4. Future Industry Collaboration

13 Vendor and Procurement Strategy

Table 4: Potential SDR Vendors

Vendor	Location
National Instruments India	Bangalore
Accord Systems	Bangalore
AgileTest Technologies	Bangalore
Keysight India	Bangalore
Rohde & Schwarz India	Bangalore

Table 5: Compute Infrastructure Vendors

Vendor	Product Family
Dell India	PowerEdge
Lenovo India	ThinkSystem
HPE India	ProLiant
Supermicro Partners	Rack Servers

14 Junior Research Fellow Requirements

14.1 Essential Qualifications

- B.Tech/M.Tech in CSE/ECE/IT
- Linux Administration

- Python Programming
- Computer Networks
- Data Structures and Algorithms
- Database Management
- Version Control Systems

14.2 Desirable Qualifications

- GATE Qualification
- Open5GS Experience
- srsRAN Experience
- SDR Programming
- GNU Radio
- Wireless Communication
- AI/ML Knowledge
- Research Publication Experience

15 Risk Assessment

Risk	Impact	Mitigation
Procurement Delay	Medium	Multi-vendor sourcing
SDR Failure	Medium	Spare units
Software Bugs	High	Incremental validation
RF Interference	Medium	Shielded enclosure
Power Failure	Medium	UPS backup

Table 7: Capital Expenditure

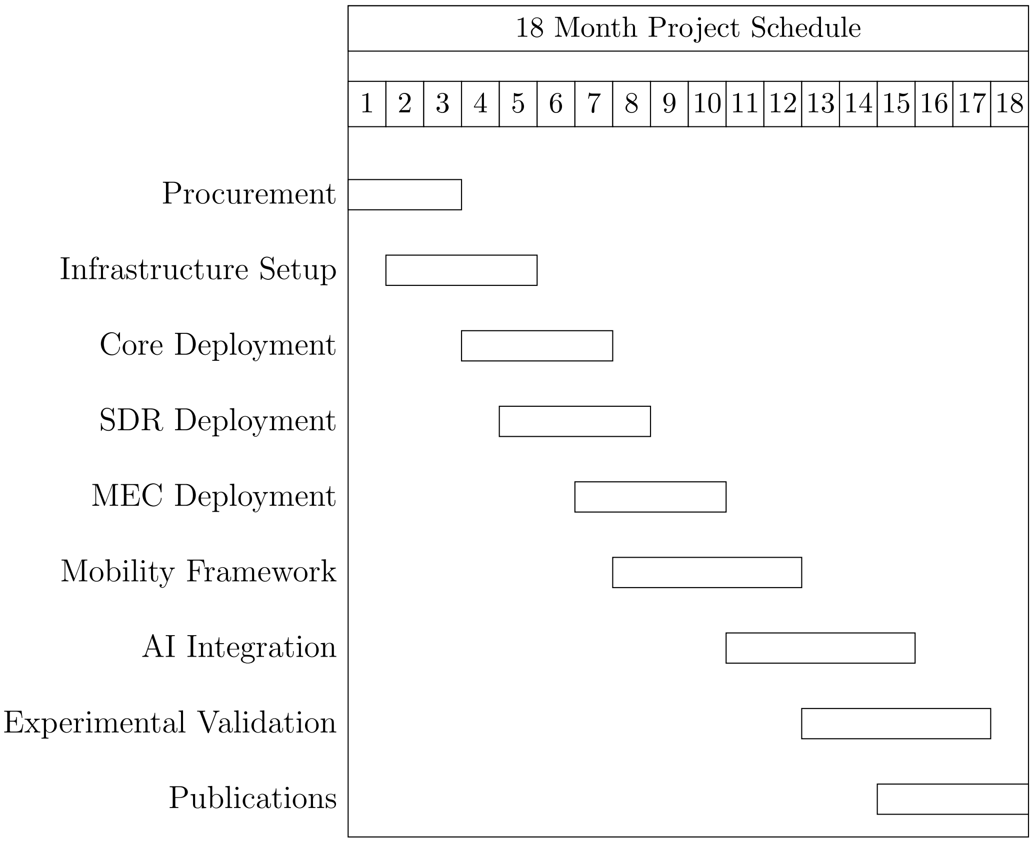
Component	Cost (INR)
Servers	8,00,000
SDRs	5,00,000
Networking	50,000
Storage	70,000
UPS	1,00,000

Table 8: Annual Operational Expenditure

Item	Cost (INR)
Electricity	60,000
Maintenance	50,000
Consumables	20,000
Software Support	30,000

16 CAPEX and OPEX Analysis

17 Project Timeline



18 Experimental Methodology

Define intra-slice mobility, inter-slice mobility, QoS preservation, congestion handling, and AI-assisted handover experiments.

19 AI Integration Roadmap

Reinforcement learning for handover decisions, traffic prediction, anomaly detection, and self-healing networks.

20 Bill of Materials

Item	Qty	Cost (INR)
Core Server	1	300000
Edge Server	1	200000
AI Workstation	1	300000
USRP B210	2	500000
Networking Equipment	1	50000
Monitoring Infrastructure	1	80000
UPS	1	100000
Storage	1	70000

21 Budget Justification

Provide technical justification for every capital expenditure item.

22 Risk Assessment

Risk	Impact	Mitigation
Hardware Delay	Medium	Alternate Vendors
SDR Failure	Medium	Spare Inventory
Configuration Complexity	High	Documentation and Training

23 Expected Deliverables

1. Real Private 5G Testbed
2. Dynamic Slice Switching Framework

3. Mobility Analytics Platform
4. Research Publications
5. Patent Opportunities